**Experiment 02:**

Study the basic network command and Network configuration commands like ping,variations of ip config,tracert,nslookup,netstat,arp,rarp,hostname,pathping and basic networking commands.

- ping
- ipconfig
- tracert
- nslookup
- netstat
- arp
- hostname
- pathping
- route
- net use

We will use a window 11 machine to understand these commands. Although some different OS like macOS & variants of Linux also follows the similar structure with different names but the working will be same.

## 1. What is Ping?

Ping is a network diagnostic command used to check whether another computer or network device is reachable over the network.

In other words,
Ping checks whether two devices can communicate with each other.

It also tells us:

- Is the destination device reachable?
- Is the network connection working?

- How much time does data take to travel?
- Were any packets lost during communication?

**What Happens When We Run the Ping Command?**

Many students think Ping simply sends a message saying "Hello."

That is **not correct**.

Ping works using a protocol called **ICMP (Internet Control Message Protocol)**.

**What is ICMP?**

**ICMP (Internet Control Message Protocol) is a network protocol used to send control, error, and diagnostic messages between network devices.**

Unlike HTTP or FTP,

ICMP is **not used for transferring files or web pages**.

It is mainly used for:

- Testing connectivity
- Reporting network errors
- Diagnosing network problems

The Ping command uses **ICMP Echo Request** and **ICMP Echo Reply** messages.

Eg: **8.8.8.8** is Google's Public DNS Server.

ping 8.8.8.8

**Ping Variations:**

1. ping -t:

**The ping -t command sends ICMP Echo Request packets continuously until the user manually stops the command.**

In simple words,

**ping -t keeps checking whether the destination device is reachable until you stop it yourself.**

```
ping -t google.com
```

Press ctrl+C to stop.

```
C:\Users\Lakshdeep Kashyap>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=13ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114
Reply from 8.8.8.8: bytes=32 time=21ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=16ms TTL=114
Reply from 8.8.8.8: bytes=32 time=17ms TTL=114
Reply from 8.8.8.8: bytes=32 time=41ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114
Reply from 8.8.8.8: bytes=32 time=15ms TTL=114
Reply from 8.8.8.8: bytes=32 time=22ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=16ms TTL=114
Reply from 8.8.8.8: bytes=32 time=201ms TTL=114
Reply from 8.8.8.8: bytes=32 time=32ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 15, Received = 15, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 201ms, Average = 32ms
Control-C
^C
```

## 2. ping -n:

**The ping -n command is used to specify the number of ICMP Echo Request packets that should be sent to the destination.**

In simple words,

**ping -n allows us to decide how many Ping requests Windows should send.**

```
ping -n 20 192.168.1.1
```

```
C:\Users\Lakshdeep Kashyap>ping -n 2 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=30ms TTL=114
Reply from 8.8.8.8: bytes=32 time=25ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 30ms, Average = 27ms
```

## 3. ping -l:

The -l option is used to specify the size of the data payload (in bytes) that is sent in each ICMP Echo Request packet.

```
ping -l 100 google.com
```

```
C:\Users\Lakshdeep Kashyap>ping -l 8 8.8.8.8

Pinging 8.8.8.8 with 8 bytes of data:
Reply from 8.8.8.8: bytes=8 time=16ms TTL=114
Reply from 8.8.8.8: bytes=8 time=34ms TTL=114
Reply from 8.8.8.8: bytes=8 time=11ms TTL=114
Reply from 8.8.8.8: bytes=8 time=16ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 34ms, Average = 19ms
```

## 4. ping -f:

Fragmentation is the process of dividing a large IP packet into smaller packets so that it can travel through a network with a smaller Maximum Transmission Unit (MTU).

MTU tells us the maximum packet size a network can carry in one piece.

The letter **f** stands for:

**Don't Fragment**

It tells Windows:

**"Send the packet, but do NOT allow it to be fragmented."**

The -f option sets the "Don't Fragment (DF)" flag in the IP header. If the packet is too large for any link along the path, it is discarded instead of being fragmented.

```
ping -f -l 1472 google.com
```

```
C:\Users\Lakshdeep Kashyap>ping -f -l 8 8.8.8.8

Pinging 8.8.8.8 with 8 bytes of data:
Reply from 8.8.8.8: bytes=8 time=15ms TTL=114
Reply from 8.8.8.8: bytes=8 time=17ms TTL=114
Reply from 8.8.8.8: bytes=8 time=62ms TTL=114
Reply from 8.8.8.8: bytes=8 time=12ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 62ms, Average = 26ms
```

## 5. ping -a:

**The -a option resolves an IP address into its hostname (computer name), if the name can be resolved.**

In simple English,

**ping -a tries to find the name of the computer from its IP address.**

```
ping -a 192.168.1.10
```

```
C:\Users\Lakshdeep Kashyap>ping -a 8.8.8.8

Pinging dns.google [8.8.8.8] with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=20ms TTL=114
Reply from 8.8.8.8: bytes=32 time=12ms TTL=114
Reply from 8.8.8.8: bytes=32 time=15ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 20ms, Average = 14ms
```

# 2. What is IPCONFIG?

**IPCONFIG (Internet Protocol Configuration) is a Windows command-line utility used to view and manage the network configuration of a computer.**

1. ipconfig

```
C:\Users\Lakshdeep Kashyap>ipconfig

Windows IP Configuration


Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::f853:9fed:1777:e82a%37
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9d6b:7fd:f50b:82f8%54
   IPv4 Address. . . . . . . . . . . : 172.20.208.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : cuchdit.in
   IPv4 Address. . . . . . . . . . . : 172.25.177.37
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.25.176.1

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.121.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.137.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

2. Ipconfig /all

ipconfig /all displays the complete TCP/IP configuration of all network adapters installed on the computer.

```
C:\Users\Lakshdeep Kashyap>ipconfig -all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-IKJAHD2F
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : cuchdit.in

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Hyper-V Virtual Ethernet Adapter
   Physical Address. . . . . . . . . : 00-15-5D-AA-D4-68
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::f853:9fed:1777:e82a%37(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.18.224.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 620762461
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-55-85-AF-E8-9C-25-1D-60-85
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Hyper-V Virtual Ethernet Adapter #2
   Physical Address. . . . . . . . . : 00-15-5D-9A-19-85
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::9d6b:7fd:f50b:82f8%54(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.20.208.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 905975133
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-55-85-AF-E8-9C-25-1D-60-85
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
   Physical Address. . . . . . . . . : FA-54-F6-20-87-A9
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
   Physical Address. . . . . . . . . : FA-54-F6-20-87-B9
```

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : cuchdit.in
   Description . . . . . . . . . . . : MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
   Physical Address. . . . . . . . . : F8-54-F6-20-87-89
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 172.25.177.37(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Lease Obtained. . . . . . . . . . : 15 July 2026 10:37:14
   Lease Expires . . . . . . . . . . : 15 July 2026 12:07:46
   Default Gateway . . . . . . . . . : 172.25.176.1
   DHCP Server . . . . . . . . . . . : 172.19.2.120
   DNS Servers . . . . . . . . . . . : 172.19.2.101
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.121.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 15 July 2026 10:37:37
   Lease Expires . . . . . . . . . . : 15 July 2026 11:22:47
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.121.254
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.137.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 15 July 2026 10:37:40
   Lease Expires . . . . . . . . . . : 15 July 2026 11:22:47
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.137.254
   Primary WINS Server . . . . . . . : 192.168.137.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : cuchdit.in
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : E8-9C-25-1D-60-85
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

3. ipconfig /release

ipconfig /release releases (returns) the current IPv4 address obtained from the DHCP server.

```
C:\Users\Lakshdeep Kashyap>ipconfig -release

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Ethernet while it has its media disconnected.

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::f853:9fed:1777:e82a%37
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9d6b:7fd:f50b:82f8%54
   IPv4 Address. . . . . . . . . . . : 172.20.208.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : cuchdit.in
```

4. ipconfig /renew:

It requests a **new IP address from the DHCP server**, and together these two commands form one of the most common troubleshooting procedures used by network engineers.

```
C:\Users\Lakshdeep Kashyap>ipconfig -renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Ethernet while it has its media disconnected.

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8638:7c0b:e428:9da1%37
   IPv4 Address. . . . . . . . . . . : 172.18.224.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9d6b:7fd:f50b:82f8%54
   IPv4 Address. . . . . . . . . . . : 172.20.208.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : cuchdit.in
   IPv4 Address. . . . . . . . . . . : 172.25.181.157
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.25.176.1

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.121.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.137.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

## 3. What is Tracert?

Imagine you are opening www.google.com, but the website is loading very slowly.

You may ask:

- Is my Wi-Fi working properly?
- Is my ISP (Internet Service Provider) having a problem?
- At which router is the delay occurring?
- Is the destination server reachable?
- Through which routers is my data travelling?

A simple ping command only tells us:

- Whether the destination is reachable.
- The response time.

But it **does not tell us the path** taken by the data.

This is where the **tracert** command becomes useful.

It shows **every router (hop)** through which the packet travels before reaching the destination.

tracert stands for **Trace Route**

It is a Windows command used to trace the complete path taken by data packets from the source computer to the destination computer.

It displays:

- Every intermediate router (Hop)
- Response time of each hop
- IP address or hostname of each router
- Whether any router is delaying or dropping packets

Example:

```cmd
tracert 8.8.8.8
```

Output:

```
C:\Users\Lakshdeep Kashyap>tracert -h 5 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 5 hops:

  1     4 ms     3 ms     5 ms  172.25.176.1
  2     4 ms     3 ms     3 ms  192.168.50.11
  3     7 ms     4 ms     4 ms  192.168.50.1
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.

Trace complete.
```

On **Windows**:

- tracert uses **ICMP Echo Request** packets with increasing TTL values.

On many **Linux/Unix** systems:

- traceroute traditionally uses **UDP** packets by default (though it can also use ICMP or TCP depending on options).

**Variations of tracert:**

1. tracert -d destination: Does not resolve host names. Shows only IP addresses, making the command faster.

2. tracert -h <max_hops> destination: Specifies the maximum number of hops to search.

```
C:\Users\Lakshdeep Kashyap>tracert -h 5 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 5 hops:

  1     4 ms     3 ms     5 ms  172.25.176.1
  2     4 ms     3 ms     3 ms  192.168.50.11
  3     7 ms     4 ms     4 ms  192.168.50.1
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.

Trace complete.
```

3. tracert -?: Displays help for the tracert command.

## 4. What is nslookup?

Many of us know  websites by names like:

- www.google.com
- www.youtube.com
- www.amazon.in

But computers **do not communicate using names**. They communicate using **IP addresses**.

The job of **DNS** is to convert a **domain name into an IP address** (and sometimes an IP address back into a domain name).

The nslookup command allows us to check whether the DNS server is working correctly and to find the IP address of a domain name.

### What is the requirement of nslookup?

Imagine you type:

www.google.com

into your web browser.

Your browser cannot directly communicate with the name **google.com**.

It first asks a **DNS Server**:

"What is the IP address of google.com?"

The DNS server replies with something like:

142.250.183.238

Only after getting the IP address does your computer connect to Google's server.

If the DNS server is not working:

- Websites may not open.
- Internet may appear to be down, even if the network connection is fine.
- You may be able to ping an IP address, but not a website name.

This is why the nslookup command is very useful.

Example:

cmd

```
nslookup google.com
```

Output:

```
C:\Users\Lakshdeep Kashyap>nslookup google.com
Server:  DC2K16.cuchdit.in
Address:  172.19.2.101

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:101f::66
          2404:6800:4000:101f::64
          2404:6800:4000:101f::8a
          2404:6800:4000:101f::8b
          142.250.118.138
          142.250.118.139
          142.250.118.100
          142.250.118.113
          142.250.118.101
          142.250.118.102
```

Non-Authoritative Answer means, The DNS server is giving the answer from its **cache**, not directly from the original authoritative DNS server for that domain.

Addresses: 142.250.183.238 (IPv4)
2404:6800:4007:80b::200e (IPv6)

| Command | Purpose | Example |
|---|---|---|
| nslookup google.com | Find the IP address of a webs | nslookup google.com |
| nslookup 8.8.8.8 | Reverse lookup (IP to doma | nslookup 8.8.8.8 |
| nslookup google.com 8.8.8.8 | Use a specific DNS server | nslookup google.com 8.8.8.8 |
| nslookup -type=A google.com | Show IPv4 records | nslookup -type=A google.com |
| nslookup -type=AAAA google.com | Show IPv6 records | nslookup -type=AAAA google.com |

## 5. What is netstat?

netstat stands for **Network Statistics**

It is a command-line utility used to display information about:

- Active network connections
- Listening ports
- Routing information
- Network statistics
- Protocol statistics

Network administrators use this command to monitor network activity and troubleshoot network-related problems

### Why Do We Need netstat?

Imagine your Internet is running very slowly.

Possible reasons:

- Too many applications are using the Internet.
- A malicious program is secretly connected to another computer.
- A service is listening on an unwanted port.
- An application has too many open connections.

Instead of guessing,

netstat allows us to see all current network activity.

Example:

```cmd
netstat
```

Output:

```
C:\Users\Lakshdeep Kashyap>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:1042         3ca52znvmj:57106       ESTABLISHED
  TCP    127.0.0.1:1042         3ca52znvmj:59309       ESTABLISHED
  TCP    127.0.0.1:9012         3ca52znvmj:50064       ESTABLISHED
  TCP    127.0.0.1:13030        3ca52znvmj:49670       ESTABLISHED
  TCP    127.0.0.1:13031        3ca52znvmj:49706       ESTABLISHED
  TCP    127.0.0.1:49670        3ca52znvmj:13030       ESTABLISHED
  TCP    127.0.0.1:49671        3ca52znvmj:49672       ESTABLISHED
```

| Command | Purpose | Example |
|---|---|---|
| netstat | Show active TCP connections | netstat |
| netstat -a | Show all connections and listening ports | netstat -a |
| netstat -n | Show numeric IP addresses and port numbers | netstat -n |
| netstat -o | Show Process IDs (PID) | netstat -o |
| netstat -e | Show Ethernet statistics | netstat -e |
| netstat -r | Show routing table | netstat -r |
| netstat -p TCP | Show only TCP connections | netstat -p TCP |
| netstat -p UDP | Show only UDP connections | netstat -p UDP |
| netstat -s | Show protocol statistics | netstat -s |

| | | |
|---|---|---|
| netstat -ano | Show all connections with numeric addresses and PID | netstat -ano |
| netstat -? | Show help | netstat -? |

```
C:\Users\Lakshdeep Kashyap>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:445            LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:902            LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:912            LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:2179           LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:3306           LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:5040           LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:5357           LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:5432           LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:12177          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:33060          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49664          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49665          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49666          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49667          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49668          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49680          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49681          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49682          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49683          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49685          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49686          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49687          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49688          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49689          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    0.0.0.0:49705          LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:1042         LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:1042         3ca52znvmj:57106       ESTABLISHED
  TCP    127.0.0.1:1042         3ca52znvmj:59309       ESTABLISHED
  TCP    127.0.0.1:1043         LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:7778         LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:9012         LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:9012         3ca52znvmj:50064       ESTABLISHED
  TCP    127.0.0.1:9013         LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:11001        LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:13030        LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:13030        3ca52znvmj:49670       ESTABLISHED
  TCP    127.0.0.1:13031        LAPTOP-IKJAHD2F:0      LISTENING
  TCP    127.0.0.1:13031        3ca52znvmj:49706       ESTABLISHED
  TCP    127.0.0.1:13032        LAPTOP-IKJAHD2F:0      LISTENING
```

```
C:\Users\Lakshdeep Kashyap>netstat -o

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    127.0.0.1:1042         3ca52znvmj:57106       ESTABLISHED     7452
  TCP    127.0.0.1:1042         3ca52znvmj:59309       ESTABLISHED     7452
  TCP    127.0.0.1:9012         3ca52znvmj:50064       ESTABLISHED     15196
  TCP    127.0.0.1:13030        3ca52znvmj:49670       ESTABLISHED     5468
  TCP    127.0.0.1:13031        3ca52znvmj:49706       ESTABLISHED     4956
  TCP    127.0.0.1:49670        3ca52znvmj:13030       ESTABLISHED     5468
  TCP    127.0.0.1:49671        3ca52znvmj:49672       ESTABLISHED     7616
  TCP    127.0.0.1:49672        3ca52znvmj:49671       ESTABLISHED     7616
  TCP    127.0.0.1:49673        3ca52znvmj:49674       ESTABLISHED     7616
  TCP    127.0.0.1:49674        3ca52znvmj:49673       ESTABLISHED     7616
  TCP    127.0.0.1:49706        3ca52znvmj:13031       ESTABLISHED     4916
  TCP    127.0.0.1:50064        3ca52znvmj:9012        ESTABLISHED     2724
  TCP    127.0.0.1:50100        3ca52znvmj:50618       ESTABLISHED     4956
  TCP    127.0.0.1:50618        3ca52znvmj:50100       ESTABLISHED     2724
  TCP    127.0.0.1:57106        3ca52znvmj:1042        ESTABLISHED     12096
  TCP    127.0.0.1:59309        3ca52znvmj:1042        ESTABLISHED     15196
```

```
C:\Users\Lakshdeep Kashyap>netstat -e
Interface Statistics

                           Received            Sent

Bytes                    3365241186      3034383672
Unicast packets            56637916        11601772
Non-unicast packets         1438504           73010
Discards                          0               0
Errors                            0               0
Unknown protocols                 0
```

```
C:\Users\Lakshdeep Kashyap>netstat -r
===========================================================================
Interface List
 37...00 15 5d fe f0 71 ......Hyper-V Virtual Ethernet Adapter
 54...00 15 5d ce 3d 9c ......Hyper-V Virtual Ethernet Adapter #2
 18...fa 54 f6 20 87 a9 ......Microsoft Wi-Fi Direct Virtual Adapter #3
  3...fa 54 f6 20 87 b9 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 12...f8 54 f6 20 87 89 ......MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
 10...00 50 56 c0 00 01 ......VMware Virtual Ethernet Adapter for VMnet1
 22...00 50 56 c0 00 08 ......VMware Virtual Ethernet Adapter for VMnet8
 15...e8 9c 25 1d 60 85 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0     172.25.176.1   172.25.181.157     30
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     172.18.224.0    255.255.240.0         On-link      172.18.224.1    271
     172.18.224.1  255.255.255.255         On-link      172.18.224.1    271
   172.18.239.255  255.255.255.255         On-link      172.18.224.1    271
     172.20.208.0    255.255.240.0         On-link      172.20.208.1    271
     172.20.208.1  255.255.255.255         On-link      172.20.208.1    271
   172.20.223.255  255.255.255.255         On-link      172.20.208.1    271
     172.25.176.0    255.255.240.0         On-link    172.25.181.157    286
   172.25.181.157  255.255.255.255         On-link    172.25.181.157    286
   172.25.191.255  255.255.255.255         On-link    172.25.181.157    286
    192.168.121.0    255.255.255.0         On-link     192.168.121.1    291
    192.168.121.1  255.255.255.255         On-link     192.168.121.1    291
  192.168.121.255  255.255.255.255         On-link     192.168.121.1    291
    192.168.137.0    255.255.255.0         On-link     192.168.137.1    291
    192.168.137.1  255.255.255.255         On-link     192.168.137.1    291
  192.168.137.255  255.255.255.255         On-link     192.168.137.1    291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     192.168.121.1    291
        224.0.0.0        240.0.0.0         On-link     192.168.137.1    291
        224.0.0.0        240.0.0.0         On-link      172.20.208.1    271
        224.0.0.0        240.0.0.0         On-link      172.18.224.1    271
        224.0.0.0        240.0.0.0         On-link    172.25.181.157    286
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     192.168.121.1    291
  255.255.255.255  255.255.255.255         On-link     192.168.137.1    291
  255.255.255.255  255.255.255.255         On-link      172.20.208.1    271
  255.255.255.255  255.255.255.255         On-link      172.18.224.1    271
  255.255.255.255  255.255.255.255         On-link    172.25.181.157    286
===========================================================================
```

## 6. What is arp?

This command is one of the most important commands because it directly relates to the **Data Link Layer (Layer 2)** and the **Network Layer (Layer 3)**.

It helps us to understand one of the most confusing concepts in networking:

**"How does a computer know the MAC Address of another computer if it only knows its IP Address?"**

The answer is:

**ARP (Address Resolution Protocol): The arp command is used to view and manage the ARP Cache, which contains mappings between IP addresses and MAC addresses.**

Suppose your computer wants to send data to another computer on the same LAN.

Your computer knows:

Destination IP Address: 192.168.0.50

But Ethernet communication cannot happen using only an IP address.

Ethernet frames require:

Destination MAC Address

Now the question is:

**How will the sender know the MAC Address corresponding to 192.168.0.50?**

The answer is:

Using **ARP**.

The arp command allows us to view and manage this IP-to-MAC mapping stored on our computer.

Examples:

```cmd
arp -a
```

Display all the entries stored inside the ARP Cache.

Output:

```
C:\Users\Lakshdeep Kashyap>arp -a

Interface: 192.168.121.1 --- 0xa
  Internet Address      Physical Address      Type
  192.168.121.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.25.181.157 --- 0xc
  Internet Address      Physical Address      Type
  172.25.176.1          cc-5a-53-d3-d5-68     dynamic
  172.25.191.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.137.1 --- 0x16
  Internet Address      Physical Address      Type
  192.168.137.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.18.224.1 --- 0x25
  Internet Address      Physical Address      Type
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.20.208.1 --- 0x36
  Internet Address      Physical Address      Type
  172.20.223.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

What does this 0x8 means?

**Interface Index**

Windows assigns every network adapter a unique identification number.

Example

| Interface | Interface Index |
|---|---|
| Wi-Fi | 0x8 |
| Ethernet | 0x9 |
| VPN | 0x12 |

cmd

```
arp -d 192.168.0.15
```

Delete an entry from the ARP Cache.

cmd

```
arp -d *
```

Delete all entries

cmd

```
arp -s 192.168.0.15 00-15-5D-45-12-AB
```

Create a **Static ARP Entry**.

Means, windows permanently stores:
192.168.0.15

↓

00-15-5D-45-12-AB

instead of learning it automatically.

Administrator privilege is required.

## 7. What is hostname?

**The hostname command is used to display the name of the current computer.**

How top change hostname: press wind + r -> type sysdm.cpl

```
C:\Users\Lakshdeep Kashyap>hostname
LAPTOP-IKJAHD2F
```

Over the network we follow a order:

**Useful Tip**

Order of commands;

hostname

Shows **Who am I?**

ipconfig

Shows **What is my IP Address?**

arp -a

Shows **What MAC Addresses have I learned on my local network?**

## 8. What is pathping?

Pathping = Ping + Tracert

It combines the features of both commands.

Suppose a user complains:

"Google opens very slowly."

Now, you need to answer:

- Is the destination reachable?
- Which route is the packet taking?
- Which router is causing the delay?
- At which router are packets being lost?

Neither ping nor tracert alone can answer all these questions.

The pathping command provides all this information in a single command.

**Path Ping**

It is a Windows networking command that combines the functionality of:

- ping
- tracert

It traces the route to the destination and then sends multiple ping packets to **each router (hop)** to calculate:

- Packet Loss
- Response Time
- Network Delay

In simple terms, **The pathping command traces the path to a destination and analyzes packet loss and latency at each router along the route.**

**Why Do We Need pathping?**

Imagine a route like this:

Your Laptop

|

▼

Home Router

|

▼

ISP Router

|

▼

City Router

|

▼

Google Server

Suppose the **ISP Router** is overloaded.

Result:

- Some packets are dropped.
- Internet becomes slow.

A normal ping only tells you:

Reply from Google

A tracert only tells you:

The packet passed through ISP Router.

But **pathping** tells you:

Packet Loss at ISP Router = 18%

Now you know exactly where the problem is.

‹/› cmd

```
pathping google.com
```

Output:

```
C:\Users\Lakshdeep Kashyap>pathping 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:
  0  LAPTOP-IKJAHD2F.cuchdit.in [172.25.181.157]
  1  172.25.176.1
  2  192.168.50.11
  3  192.168.50.1
  4     *        *        *
Computing statistics for 75 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           LAPTOP-IKJAHD2F.cuchdit.in [172.25.181.157]
                                0/ 100 =  0%   |
  1    9ms     0/ 100 =  0%     0/ 100 =  0%  172.25.176.1
                                0/ 100 =  0%   |
  2   10ms     0/ 100 =  0%     0/ 100 =  0%  192.168.50.11
                                0/ 100 =  0%   |
  3    9ms     0/ 100 =  0%     0/ 100 =  0%  192.168.50.1

Trace complete.
```

Each hop represents one router between the source and the destination.

Example:

| Hop | Device |
|---|---|
| 0 | Your Computer |
| 1 | Home Router |
| 2 | ISP Router |
| 3 | Internet Router |
| 4 | Google Server |

**Lost/Sent (Source to Here)**

Example

2/100

Meaning:

Out of **100 packets sent**, **2 packets were lost** before reaching this hop.

Packet Loss:

2%

**Lost/Sent (This Node/Link)**

Example

3/100

This tells how many packets were lost **at that specific router or link**.

This helps identify the exact location of the problem.

**Address**

Shows the IP address or Host Name of the router.

Example

192.168.0.1

Your Home Router.

---

10.20.0.1

ISP Router.

---

google.com

Destination Server.

**What is Packet Loss?**

Packet Loss means that some packets sent by your computer never reached their destination.

Example:

You send:

100 Packets

Destination receives:

95 Packets

Then,

Packet Loss = 5%

**Formula**

Packet Loss (%) = (Packets Lost / Packets Sent) × 100

**Example**

Packets Sent:

100

Packets Lost:

5

Calculation:

Packet Loss = (5 / 100) × 100

Packet Loss = 5%

**Why Does Packet Loss Occur?**

Common reasons:

- Network congestion.
- Faulty cables.
- Weak Wi-Fi signal.
- Router overload.
- Hardware failure.
- ISP issues.

---

**Which Protocol Does pathping Use?**

pathping uses **ICMP (Internet Control Message Protocol)**.

It sends:

- ICMP Echo Requests.
- ICMP Echo Replies.

Just like ping, but it also traces the route like tracert.

**Practical Uses**

Network administrators use pathping to:

- Detect packet loss.
- Identify slow routers.
- Troubleshoot network performance.
- Verify routing paths.

- Diagnose ISP-related issues.